

Sample Case study on class & objects CO1

Bank Account Management System:

Problem Statement: Design a system to manage bank accounts.

Classes: Account, Savings-Account, Current-Account.

Attributes: Account number, balance, account type, interest rate (for savings account), overdraft limit (for current account), etc.

Methods: Deposit, Withdraw, Calculate Interest, Transfer, etc.

Object Usage: Create objects for each customer's account and perform transactions like

deposits, withdrawals, and transfers.

Bank Management System

Algorithm: Bank Management System

Step 1: Start the program.

Step 2: Create a Scanner object.

Step 3: Create a SavingsAccount object with:

- Account number = S101
- Initial balance = 10000
- Interest rate = 5%

Step 4: Create a CurrentAccount object with:

- Account number = C101
- Initial balance = 5000
- Overdraft limit = 5000

Step 5: Display the details of the Savings Account.

Step 6: Perform operations on the Savings Account:

1. Deposit 2000.
2. Withdraw 1000.
3. Calculate and add interest.

4. Display the interest added.

Step 7: Display the details of the Current Account.

Step 8: Perform operations on the Current Account:

1. Deposit 3000.
2. Withdraw 10000.
3. Check whether the withdrawal is within the balance plus overdraft limit.

Step 9: Transfer 2000 from the Savings Account to the Current Account.

Step 10: Display the final Savings Account details.

Step 11: Display the final Current Account details.

Step 12: Close the Scanner.

Step 13: Stop the program.

2. Execution Steps

Step 1: Create the Java Project

Open **Eclipse IDE** and create a Java project.

Create the package:

package observation;

Step 2: Create the Classes

The program contains the following classes:

1. Account
2. SavingsAccount
3. CurrentAccount
4. BankManagement

Step 3: Compile and Run

Save the program and select:

Run → Run As → Java Application

Step 4: Savings Account Operations

The Savings Account is initially created with:

Account No: S101

Account Type: Savings

Balance: 10000.0

Then:

Deposit: 2000

Balance = 12000

Withdraw:

Withdrawal: 1000

Balance = 11000

Interest calculation:

Interest = $11000 \times 5 / 100$

= 550

Therefore:

Final Savings Balance = 11550

Step 5: Current Account Operations

Initially:

Account No: C101

Account Type: Current

Balance: 5000.0

After depositing 3000:

Balance = 8000

Then withdrawal of 10000 is performed.

The overdraft limit is 5000, so the maximum allowed withdrawal is:

$8000 + 5000 = 13000$

Therefore, the withdrawal is successful.

New balance:

$8000 - 10000 = -2000$

Step 6: Transfer Operation

2000 is transferred from Savings Account to Current Account.

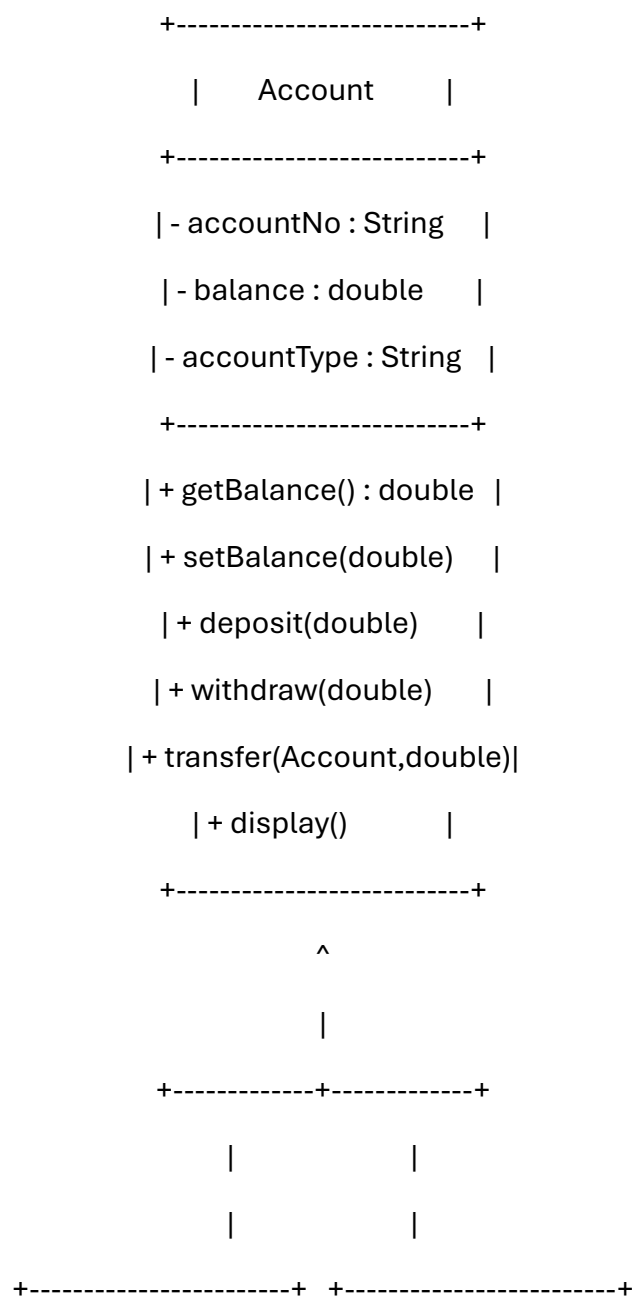
Savings:

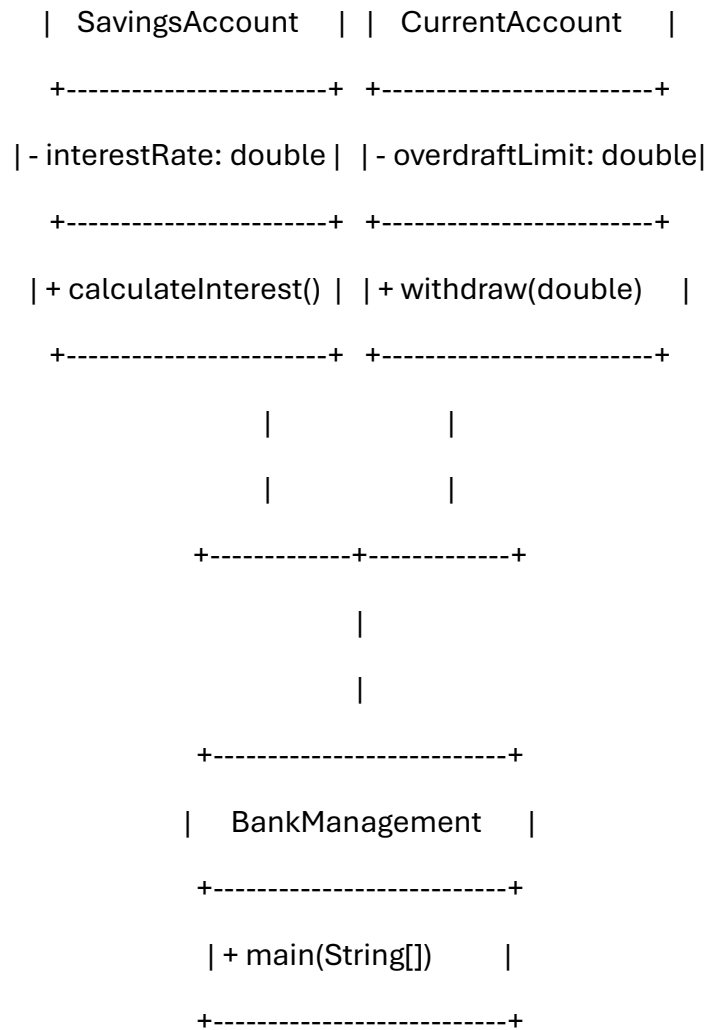
$$11550 - 2000 = 9550$$

Current:

$$-2000 + 2000 = 0$$

4.UML Class Diagram





Source Code:

```
package observation;
```

```
import java.util.Scanner;
```

```
class Account {
    private String accountNo;
    private double balance;
    private String accountType;
```

```
Account(String accountNo, double balance, String accountType) {  
    this.accountNo = accountNo;  
    this.balance = balance;  
    this.accountType = accountType;  
}
```

```
public double getBalance() {  
    return balance;  
}
```

```
public void setBalance(double balance) {  
    this.balance = balance;  
}
```

```
public void deposit(double amount) {  
    balance += amount;  
    System.out.println("Deposit successful.");  
}
```

```
public void withdraw(double amount) {  
    if (amount <= balance) {  
        balance -= amount;  
        System.out.println("Withdrawal successful.");  
    } else {  
        System.out.println("Insufficient balance.");  
    }  
}
```

```
public void transfer(Account receiver, double amount) {  
    if (amount <= balance) {  
        balance -= amount;  
        receiver.balance += amount;  
        System.out.println("Transfer successful.");  
    } else {  
        System.out.println("Insufficient balance.");  
    }  
}
```

```
public void display() {  
    System.out.println("Account No: " + accountNo);  
    System.out.println("Account Type: " + accountType);  
    System.out.println("Balance: " + balance);  
}  
}
```

```
class SavingsAccount extends Account {  
    private double interestRate;  
  
    SavingsAccount(String no, double balance, double rate) {  
        super(no, balance, "Savings");  
        interestRate = rate;  
    }  
  
    public void calculateInterest() {  
        double interest = getBalance() * interestRate / 100;  
        setBalance(getBalance() + interest);  
    }  
}
```

```
        System.out.println("Interest added: " + interest);
    }
}
```

```
class CurrentAccount extends Account {
    private double overdraftLimit;

    CurrentAccount(String no, double balance, double limit) {
        super(no, balance, "Current");
        overdraftLimit = limit;
    }
}
```

```
@Override
public void withdraw(double amount) {
    if (amount <= getBalance() + overdraftLimit) {
        setBalance(getBalance() - amount);
        System.out.println("Withdrawal successful.");
    } else {
        System.out.println("Overdraft limit exceeded.");
    }
}
}
```

```
public class BankManagement {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
```

```
SavingsAccount savings =
```

```
    new SavingsAccount("S101", 10000, 5);
```

```
CurrentAccount current =
```

```
    new CurrentAccount("C101", 5000, 5000);
```

```
System.out.println("----- SAVINGS ACCOUNT -----");
```

```
savings.display();
```

```
savings.deposit(2000);
```

```
savings.withdraw(1000);
```

```
savings.calculateInterest();
```

```
System.out.println("\n----- CURRENT ACCOUNT -----");
```

```
current.display();
```

```
current.deposit(3000);
```

```
current.withdraw(10000);
```

```
System.out.println("\n----- TRANSFER -----");
```

```
savings.transfer(current, 2000);
```

```
System.out.println("\nFinal Savings Account:");
```

```
savings.display();
```

```
System.out.println("\nFinal Current Account:");
```

```
current.display();
```

```
        sc.close();  
    }  
}
```

Output:

```
----- SAVINGS ACCOUNT -----  
Account No: S101  
Account Type: Savings  
Balance: 10000.0  
Deposit successful.  
Withdrawal successful.  
Interest added: 550.0  
  
----- CURRENT ACCOUNT -----  
Account No: C101  
Account Type: Current  
Balance: 5000.0  
Deposit successful.  
Withdrawal successful.  
  
----- TRANSFER -----  
Transfer successful.  
|  
Final Savings Account:  
Account No: S101  
Account Type: Savings  
Balance: 9550.0  
  
Final Current Account:  
Account No: C101  
Account Type: Current  
Balance: 0.0
```